



NewBear Computing Store Ltd



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Design Note 53

Nov. 78

77-68 32K - DYNARAM

This PCB accomodates two 16K bytes blocks of dynamic RAM, holds all the refresh control logic and provides 16 nominal data rates up to 4800 baud. *a clock generator for*

The following signals are available on the 77-68 bus and form *used to interface the memory board to the* the interface with the CPU card. *CPU board and are fully compatible with the 77/68 bus structure.*

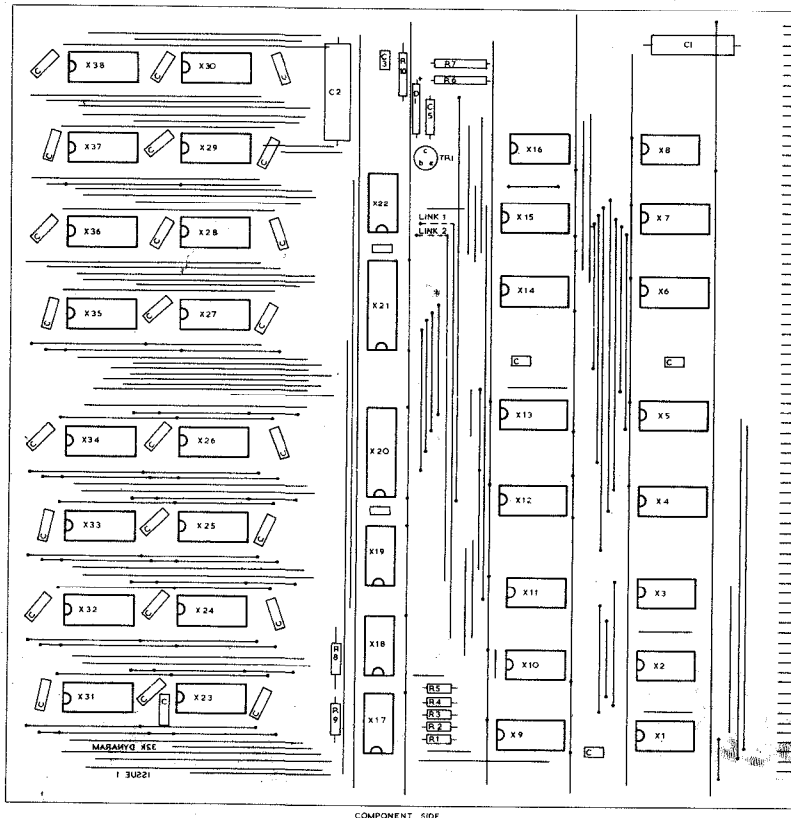
- R/W
- Ø2
- E or VMA
- 5MHz
- Voltage supplies $\pm 12V$ and 5V

The edge connector designations are compatible with the 77-68 and ROM MONITOR PCB's.

An on board 8 pole switch allows positioning of each 16K byte block anywhere within the *64 Kbyte.* 16 bit address space. An extra 5 control lines (active high) extend the address and/or allows memory to be disabled while accessing ROM or I/O anywhere within the 32K byte address space. For example this RAM could be used to 'fill-in' between MIKBUG, I/O and scratchpad locations.

Dynamic RAM's are employed with all the necessary mutliplexing and refresh logic provided - *by standard TTL L.S. chips* no special purpose chips are required. By disconnecting links 1 and 2 refresh may be exercised via the edge connectors to suit individual requirements.

When driven from the 77-68 bus the refresh cycle is 'hidden' that is it occurs during the first half of a Ø2 cycle with the advantage that there is no reduction in overall access time.



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32 kbyte 77-68 RAM BOARD